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Investigation of high Curie temperature $(1-x)\text{BiSc}_{1-y}\text{Fe}_y\text{O}_3-x\text{PbTiO}_3$ piezoelectric ceramics

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Ceramics around the morphotropic phase boundary (MPB) in the $(1-x)\text{BiSc}_{1-y}\text{Fe}_y\text{O}_3-x\text{PbTiO}_3$ solid solution were fabricated. For $y=0.5$, ceramics were single phase, and piezoelectric coefficients (d_{33}) and electromechanical coupling coefficients (k_p) for MPB compositions were 300 pC/N and 0.49, respectively; a level of piezoelectric activity similar to that of hard, lead zirconate titanate compositions but with $T_C \sim 60^\circ\text{C}$ higher at $\sim 440^\circ\text{C}$. For ceramics with $y \geq 0.7$, dielectric measurements in combination with diffraction contrast transmission electron microscopy revealed the existence of two ferroelectric phases for most PbTiO_3 contents studied. The presence of two ferroelectric phases was associated with a decrease in piezoelectric activity and although raw materials costs for $y=0.7$ and 0.8 with respect to $y=0$ were significantly lower (less Sc_2O_3) and T_C greater ($\sim 500^\circ\text{C}$), d_{33} (~ 100 pC/N) and k_p (0.18) were too low to be commercially useful for actuator applications. © 2009 American Institute of Physics. [doi:10.1063/1.3253585]

I. INTRODUCTION

Piezoelectric ceramics are used in a wide range of applications exploiting both the direct and converse piezoelectric effect. The lead zirconate-lead titanate (PZT) solid solution and its derivatives have dominated the piezoelectric ceramics market since the 1950s.^{1–4} The large intrinsic piezoelectric response of this system is attributed to the $6s^2$ electrons lone pair of Pb^{2+} that leads to considerable structural distortions from the prototype cubic phase.⁵ The large extrinsic response is attributed to a morphotropic phase boundary (MPB) composition at $\sim 48\%$ PbTiO_3 (PT) where rhombohedral and tetragonal phases coexist leading to easy movement of domain walls. A typical modified composition (PZT-5A) has $T_C \sim 365^\circ\text{C}$ and piezoelectric coefficient $d_{33} \sim 374$ pC/N (Ref. 6) but depoling effects restrict the use of these materials to temperatures below 200°C . Technological advancements in structure and control systems integration have created the need for the use of piezoelectric materials at higher temperatures in applications such as structural health monitoring and engine knock control.

In a search of such materials, various solid solutions of bismuth perovskites and lead titanate have been investigated, where the bismuth perovskite end member is either simple, $\text{Bi}(\text{Me})\text{O}_3$ ($\text{Me}^{3+}=\text{Fe}, \text{Ga}, \text{In}, \text{Mn}, \text{Sc}, \text{etc.}$), or complex, $\text{Bi}(\text{Me}'\text{Me}'')\text{O}_3$ ($\text{Me}'=\text{Mg}^{2+}, \text{Ni}^{2+}, \text{Zn}^{2+}, \text{etc.}$ and $\text{Me}''=\text{Nb}^{5+}, \text{Ti}^{4+}, \text{W}^{6+}, \text{Zr}^{4+}, \text{etc.}$). Many of these systems were reported to have high Curie temperatures combined in only a few cases with good piezoelectric response. $\text{BiFeO}_3\text{--PbTiO}_3$ (BF-PT) has long been considered for piezoelectric applications with the earliest publications dating back to the 1960s.^{7,8} It is distinctive among $\text{Bi}(\text{Me})\text{O}_3\text{--PbTiO}_3$ systems, along with $\text{Bi}(\text{Zn}_{1/2}\text{Ti}_{1/2})\text{O}_3\text{--PbTiO}_3$,⁹ for showing a continuous increase in T_C with decreasing PT content accompa-

nied by an increase in the tetragonal distortion.¹⁰ The MPB separates tilted rhombohedral (space group $R3c$) and tetragonal phases (space group $P4mm$) and the MPB composition ($\sim 30\%$ PT) has T_C of 632°C and c/a of 1.16.^{11,12} However, due to the large tetragonal distortion the application of extremely high electric fields is necessary to reorient the ferroelectric domains (coercive field is estimated to be around 150 kV/cm) which inhibits switching between the $R3c$ and $P4mm$ phases and a saturated ferroelectric hysteresis loop cannot be observed unless compositions are significantly doped.¹³ Another system, $\text{BiScO}_3\text{--PbTiO}_3$ (BS-PT) has at the MPB composition (64% PT) a T_C of 450°C , d_{33} of 460 pC/N, and strain response comparable to PZT ceramics.^{14,15} The MPB separates untilted rhombohedral (space group $R3m$) and tetragonal phases (space group $P4mm$). In the rhombohedral phase field there is an octahedral-tilt phase transition to space group $R3c$ whose onset temperature rises with decreasing PT content.¹⁶ Despite its good properties, BS-PT is not viable for most commercial applications due to the current high cost of Sc_2O_3 . A substitute for Sc_2O_3 in BS-PT is therefore desirable. Preliminary data on the $\text{Bi}(\text{Sc}_{1/2}\text{Fe}_{1/2})\text{O}_3\text{--PbTiO}_3$ system were presented and a $d_{33} \approx 300$ pC/N was reported, with T_C of 440°C .¹⁷ Here, we present a more comprehensive investigation of $(1-x)\text{BiSc}_{1-y}\text{Fe}_y\text{O}_3-x\text{PbTiO}_3$ for $y=0.5, 0.7$, and 0.8 .

II. EXPERIMENT

Ceramics in the $(1-x)\text{BiSc}_{1-y}\text{Fe}_y\text{O}_3-x\text{PbTiO}_3$ system (abbreviated to $\text{BSF}_{100y}\text{PT}_{100x}$) were prepared by the conventional solid-state reaction method. Specifically, compositions of $0.30 \leq x \leq 0.64$ with $y=0.5$, $0.425 \leq x \leq 0.55$ with $y=0.7$, and $0.45 \leq x \leq 0.50$ with $y=0.8$ were fabricated. Stoichiometric mixtures of commercial reagent-grade oxides, Bi_2O_3 , Sc_2O_3 , Fe_2O_3 , PbO , and TiO_2 , were attrition milled in distilled water with yttria-stabilized zirconia media for 1 h and dried at 100°C . The dried powders were made to pass

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through a 355 μm sieve, then reacted for 2 h at 775–900 $^{\circ}\text{C}$ depending on composition, and subsequently attrition milled as before. After drying and sieving, reacted powders were uniaxially pressed at ~ 250 MPa into green pellets and sintered for 2 h between 1000 and 1100 $^{\circ}\text{C}$. X-ray diffraction (XRD) was performed using a Siemens x-ray diffractometer with Cu $K\alpha$ radiation to ascertain the phase purity and assemblage. Scanning electron microscopy (SEM) using a JEOL 6400 microscope was performed on thermally etched pellets initially polished to 1 μm .

Samples were prepared for transmission electron microscopy (TEM) by grinding pellets to a thickness of ~ 30 μm . These were mounted with epoxy resin to copper rings and milled to electron transparency at an incidence angle of 15° using a Gatan duo ion mill (600 model). TEM was performed on a Phillips 430 TEM operating at 300 kV.

Pellets were electroded on the major pellet faces using a gold paste fired on at 800 $^{\circ}\text{C}$ for 1 h. Dielectric properties were measured up to 800 $^{\circ}\text{C}$ using an inductance/capacitance/resistance (LCR) meter (Model 4284A, Hewlett Packard). Polarization-electric field measurements were conducted using a Radiant Technologies RT-66A standardized ferroelectric test system operated on virtual ground mode at a frequency of 6 Hz. Strain-electric field measurements were performed using a noncontact fiber optic displacement probe (Philtect, 10 $\mu\text{m}/\text{V}$ sensitivity) to measure the strain. Details of the experimental setup can be found elsewhere.¹⁸ Disk samples ~ 8.5 mm in diameter and < 1 mm thick for piezoelectric measurements were poled in an oil bath under a field of 40 kV/cm at 100 $^{\circ}\text{C}$ for 5 min. The piezoelectric constant (d_{33}) was measured using a Berlincourt meter (IAAS ZJ-2). The planar electromechanical coupling factor k_p was calculated based on the resonance method using an impedance analyzer (Agilent 4294A).⁶ The variation in k_p up to 400 $^{\circ}\text{C}$ was determined by attaching the poled specimen to a conductivity jig which was placed into a noninductively wound tube furnace. The disk sample was placed between two platinum wires that were in contact (but not glued) with the electroded surfaces. The sample was supported only by the wires with its major faces horizontally positioned. The resonance spectrum was obtained with the sample held at the desired temperature after allowing sufficient time for thermal equilibrium.

III. RESULTS

Figure 1 shows the XRD patterns of different compositions for $y=0.5$, 0.7, and 0.8. All peaks were identified as perovskite and no nonperovskite, second phases were observed but as x increased the peaks within the patterns became broader. More detailed analysis of XRD patterns showed that for $y=0.5$ rhombohedral symmetry was present at room temperature for $0.45 \leq x \leq 0.50$, as evidenced by asymmetry in the $\{111\}_p$ peak. For higher PT contents, tetragonal symmetry emerged as indicated by $\{100\}_p$ peak asymmetry and the absence of $\{111\}_p$ peak splitting. The tetragonal distortion was calculated to be 1.5% and 2.4% for $x=0.53$ and 0.55, respectively. When y increased to 0.7, room temperature phases appeared rhombohedral for 0.425

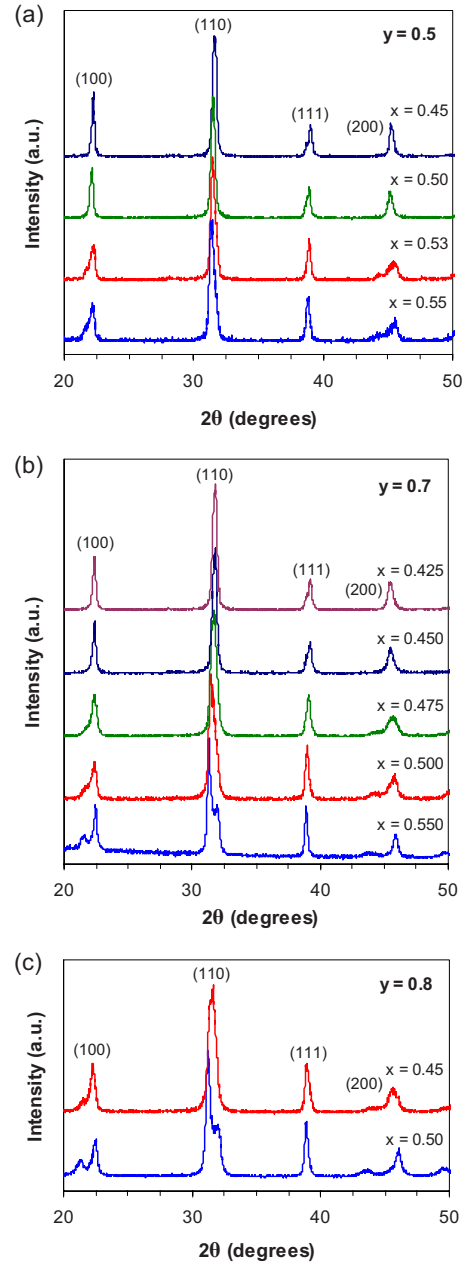


FIG. 1. (Color online) Room temperature XRD patterns of BSF-PT ceramics. (a) $y=0.5$, (b) $y=0.7$, and (c) $y=0.8$.

$\leq x \leq 0.450$ and nominally tetragonal for $0.475 \leq x \leq 0.55$ by XRD, shifting the nominal MPB to lower PT contents. When y increased further to 0.8, tetragonal symmetry was identified for $x=0.50$. For $x=0.45$, no nonperovskite phase was detected but a fit could not be established for either rhombohedral or tetragonal symmetry suggesting that both phases may coexist.

Typical grain structures of BSF-PT ceramics for different y values are shown in Fig. 2. The grain size is 0.5–2 μm , smaller than for BS-PT ceramics.¹⁵ Grain size greatly affects the extrinsic contributions to the piezoelectric performance. Fine-grained ceramics have limited domain variants and the domain reorientation is hindered by strong coupling between grain boundaries and domain walls, leading to lower polarization and piezoelectric response.^{19,20} For

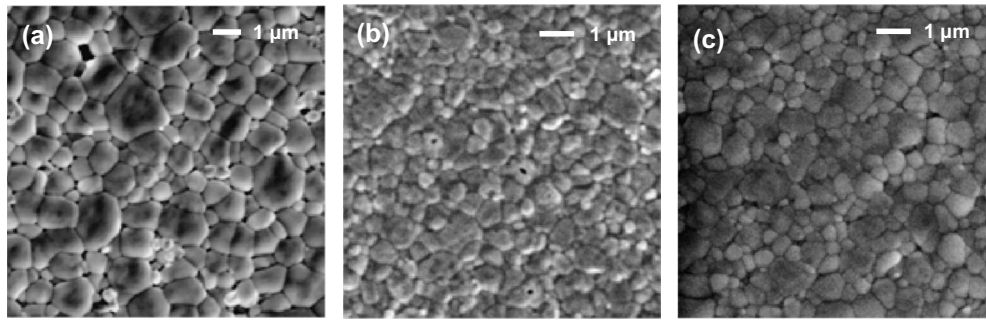


FIG. 2. SEM micrographs of polished surfaces of thermally etched pellets of (a) BSF₅₀PT53, (b) BSF₇₀PT45, and (c) BSF₈₀PT45.

Nb-doped BS-PT, the piezoelectric properties deteriorate significantly when the grain size is smaller than 1 μm .²¹

Figures 3(a)–3(c) show TEM micrographs depicting the typical domain structures of ceramics of $y=0.5$. At $x=0.50$, $y=0.5$ [Fig. 3(a)], rhombohedral at room temperature according to XRD, a highly perturbed domain structure is present with no evidence of lamellar twins. This domain structure is similar to that of rhombohedral phases of BF-PT.¹¹ Samples with $x=0.53$ [Fig. 3(b)], nominally tetragonal according to XRD with $c/a=1.5\%$ at room temperature, exhibited within one grain, regions of fine-scale lamellar domains consistent with tetragonal symmetry and regions of domain structure similar to that observed in $x=0.5$, $y=0.5$. Samples with $x=0.55$, $y=0.5$ [Fig. 3(c)] showed a qualitatively similar domain structure to $x=0.53$, $y=0.5$ but with regions of lamellar domains in greater abundance, consistent with its greater tetragonality ($c/a=2.4\%$). The diffraction contrast TEM images shown in Figs. 3(a)–3(c), qualitatively agree with the presence of a MPB for $0.5 < x < 0.55$. The coexistence of two different domain structures, one highly perturbed and the other lamellar, is characteristic of a transition from rhombohedral to tetragonal (MPB) that occurs over a relatively narrow compositional range of x ($\sim 2\text{--}3$ mol%).

In contrast, ceramics with $y=0.7$ and 0.8 did not exhibit a narrow MPB. This is exemplified by comparing the domain structures observed in BSF₇₀PT55 with BSF₇₀PT45. Accord-

ing to XRD, BSF₇₀PT55 is tetragonal and consequently exhibits a lamellar domain structure [Fig. 3(d)] in all grains. By analogy, BSF₇₀PT45 should only exhibit a domain structure consistent with a rhombohedral structure [Figs. 3(e) and 3(f)]. However, grains composed of entirely lamellar domains can still be found in BSF₇₀PT45 suggesting the intergranular coexistence of rhombohedral and tetragonal ferroelectric phases even at relatively low PT contents. In qualitative agreement therefore with the XRD data (Fig. 1) the compositional range of the MPB is greater for $y=0.7$ and 0.8 than $y=0.5$.

The relative permittivity ϵ_r and dielectric loss $\tan \delta$ for compositions with $y=0.5$, 0.7 , and 0.8 are shown in Fig. 4. ϵ_r for BSF₅₀PT55 exhibits a maximum at ~ 440 °C and although there is no temperature shift in the peak with frequency it does exhibit considerable frequency dispersion around the peak maximum probably due to high losses at high temperatures, Fig. 4(a). Figure 4(b) shows the dielectric response of BSF₇₀PT47.5, where in addition to the main peak in ϵ_r at ~ 450 °C an anomaly is observed at ~ 600 °C. This anomaly is present for all compositions where $x < 0.55$ for $y=0.7$ indicating that as the PT content is reduced two separate ferroelectric phases are present. The same effect was present for $y=0.8$, where $x=0.50$ showed a single ϵ_r

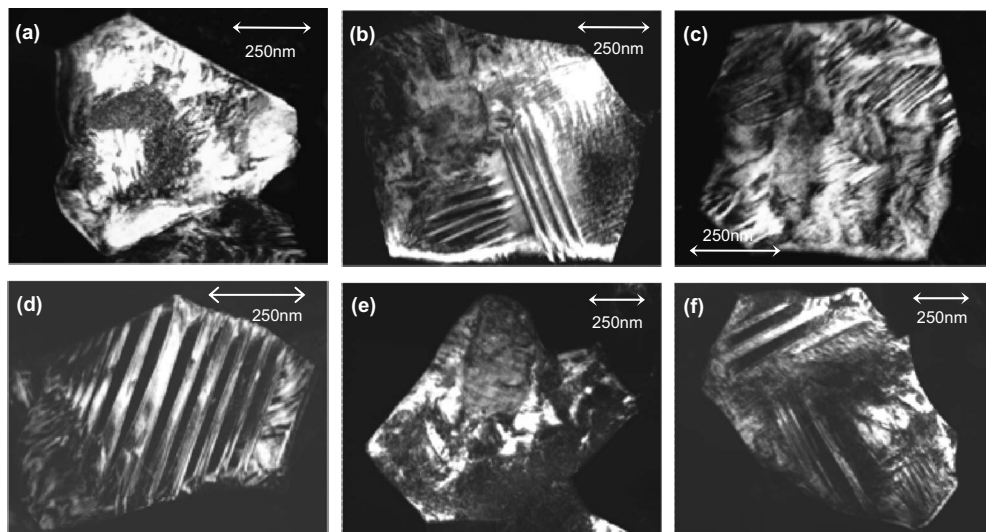


FIG. 3. Dark field TEM micrographs of the typical domain structure of (a) BSF₅₀PT50, (b) BSF₅₀PT53, (c) BSF₅₀PT55, (d) BSF₇₀PT55, and [(e) and (f)] BSF₇₀PT45 obtained along the $\langle 001 \rangle_p$ direction.

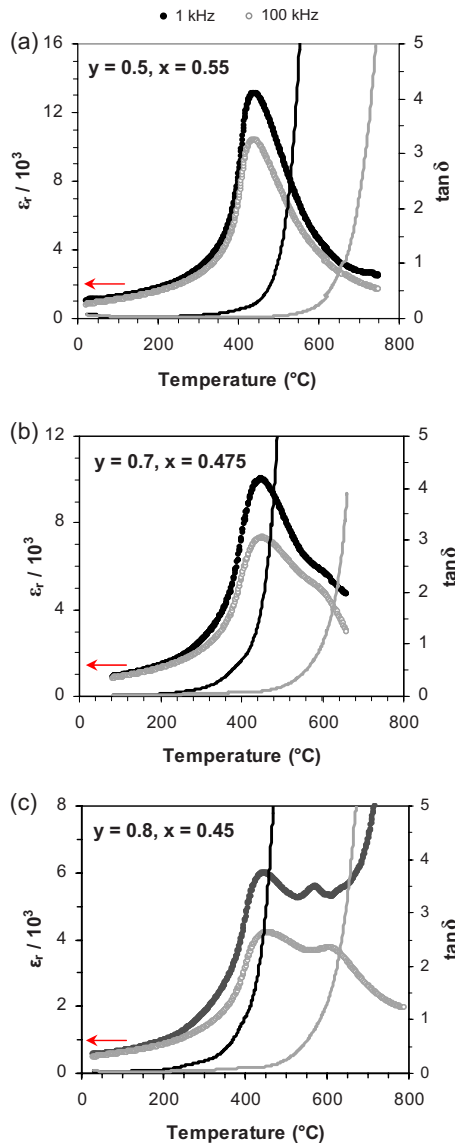


FIG. 4. (Color online) ϵ_r and $\tan \delta$ vs temperature of (a) BSF₅₀PT55, (b) BSF₇₀PT47.5, and (c) BSF₈₀PT45 measured at 1 and 100 kHz.

maximum at $\sim 500^\circ\text{C}$ (not shown here), while $x=0.45$ exhibits a maximum at $\sim 450^\circ\text{C}$ and a second local maximum at $\sim 600^\circ\text{C}$, as shown in Fig. 4(c).

Overall the data presented in Figs. 4(a)–4(c) are consistent with the premise that $y=0.5$ is essentially a single perovskite phase for compositions close to the MPB. At the MPB within a narrow compositional range, tetragonal and rhombohedral phases coexist with c/a approaching unity and rhombohedral angle close to 90° in a similar manner to PZT and BS-PT. In contrast, for $y=0.7$ and 0.8 , there is strong evidence for the coexistence of two ferroelectric phases, tetragonal and rhombohedral over a wider compositional range (~ 10 mol%) which gives rise to complex two-phase domain structures in TEM images and multiple anomalies in ϵ_r and $\tan \delta$ as a function of temperature. Compositions with $y=0.7$ and 0.8 may therefore be considered to contain a pseudo- rather than classical-MPB whose greater width results from heterogeneity.

The room temperature polarization-electric field behavior for BSF-PT ceramics with optimized piezoelectric prop-

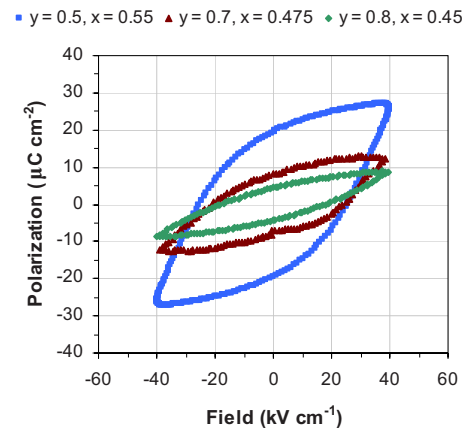


FIG. 5. (Color online) Polarization-electric field loops of BSF-PT ceramics measured at room temperature.

erties is shown in Fig. 5. The best response is shown by ceramics with $y=0.50$. The maximum polarization values decrease significantly when y increases to 0.7 and 0.8 . The strain-electric field bipolar switching behavior for BSF-PT ceramics is shown in Fig. 6(a), where the optimized response for each y value is included. The best response is shown by ceramics with $y=0.5$, $x=0.53$ with a total strain of $\sim 0.20\%$. For $y=0.7$ and 0.8 the total strains are much lower and from the shape of the loops it is evident that coercive fields are higher for these compositions. The strain-field response under unipolar drive is shown in Fig. 6(b). The maximum de-

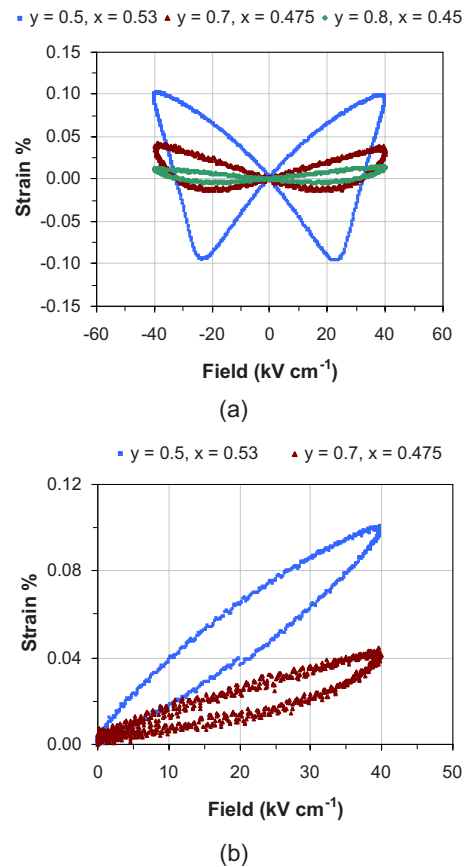


FIG. 6. (Color online) Strain-field behavior of BSF-PT ceramics under (a) bipolar and (b) unipolar drive.

TABLE I. Dielectric and piezoelectric properties of BSF-PT ceramics.

Material	T_C (°C)	ϵ_r	$\tan \delta$	d_{33} (pC/N)	k_p
BSF ₅₀ PT50	425	570	0.027	180	0.47
BSF ₅₀ PT53	428	830	0.040	290	0.50
BSF ₅₀ PT55	440	1130	0.033	298	0.49
BSF ₇₀ PT45	460	465	0.030	139	0.39
BSF ₇₀ PT475	450	749	0.032	167	0.36
BSF ₇₀ PT50	451	684	0.030	172	0.35
BSF ₈₀ PT45	454	650	0.015	150	0.31
BSF ₈₀ PT50	498	484	0.016	100	0.18

veloped unipolar strain ($\sim 0.10\%$) for $y=0.5$, $x=0.53$ is roughly half of that of BS-PT under similar driving conditions.¹⁵

Table I summarizes the piezoelectric and dielectric properties of BSF-PT ceramics. Optimum d_{33} values for $y=0.5$ and 0.7 are 300 and 170 pC/N, respectively, lower than BS-PT. Figure 7 presents d_{33} versus the temperature of maximum ϵ_r which can be estimated as the ferroelectric transition temperature. In the case of double peaks in ϵ_r the lower temperature is used. Among the compositions studied the highest d_{33} is ~ 300 pC/N with $T_C=440$ °C, and the highest attained T_C is ~ 500 °C with $d_{33}=100$ pC/N. Figure 8 shows the variation in k_p with temperature for the optimum compositions BSF₅₀PT55 and BSF₇₀PT47.5. k_p values for both samples retain their respective room temperature levels up to ~ 250 °C. The decline of k_p at higher temperatures is more rapid for BSF₅₀PT55 reaching a value of 0.2 at 325 °C in comparison to 400 °C for BSF₇₀PT47.5. These results indicate that the samples can produce piezoelectric components that can be operated at higher temperatures than PZT. Chen *et al.*²² have reported that optimized BS-PT ceramics showed only a small decline in room temperature k_p even after annealing at 400 °C, near the Curie point. These results are not directly comparable with the ones reported here since they are based on different measurement methods. The annealing experiments²² are indicative of the thermal depoling behavior of the samples but are not representative of their performance at high temperatures. Direct measurements of

the properties at high temperatures are necessary to establish the piezoelectric performance and thus the operating temperature limit.

IV. DISCUSSION

The lower piezoactivity for $y=0.7$ and 0.8 compared with $y=0.5$ can be explained by the far greater range of x over which rhombohedral and tetragonal phases coexist for higher Fe concentrations. For $y=0.5$, the narrow range of x ($\sim 2-3$ mol%) represents a classical MPB similar to PZT in which the rhombohedral and tetragonal phases are close in free energy within the majority of the ceramic. Consequently, the extrinsic contribution by domain wall motion to the piezoactivity is maximized. In the case of $y=0.7$ and 0.8, a pseudo- rather than classical-MPB is observed in the solid solution within which rhombohedral and tetragonal phases coexist. Therefore, for any given composition within the pseudo-MPB, only a small volume fraction contains rhombohedral and tetragonal phases of similar free energy and switching between the [001] and [111] polarization vectors only occurs in a small percentage of the sample. The origin of the heterogeneity at $y > 0.5$ may be either kinetic or thermodynamic. The former is a result of insufficient time at temperature to attain homogenization of the constituent ions. The latter may arise as a result of immiscibility between two ferroelectric phases at high temperature. However, it should be noted that all BSF-PT compositions were attrition milled to submicron particle size prior to reaction and sintering

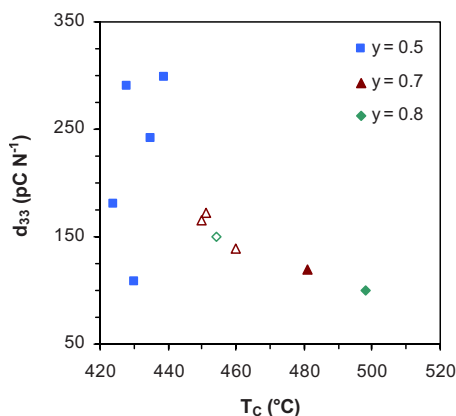


FIG. 7. (Color online) d_{33} vs Curie temperature for BSF-PT ceramics. Un-filled symbols denote samples where an additional dielectric anomaly is observed around 600 °C.

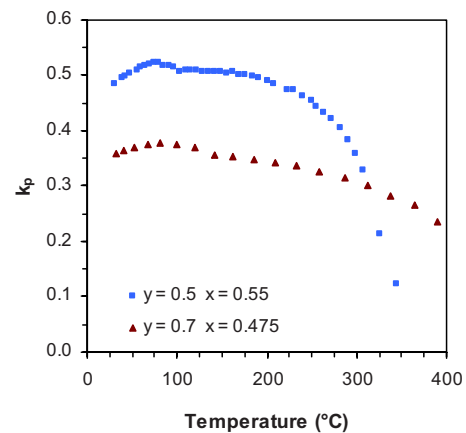


FIG. 8. (Color online) Temperature variation in k_p for BSF₅₀PT55 and BSF₇₀PT47.5 ceramics.

steps. In previous work,²³ attrition milling in the preparation of PZT-based ceramics using an identical procedure to that described here was typically sufficient to create a narrow MPB region (<2 mol%) with optimized piezoelectric coefficients within the tetragonal phase immediately adjacent to the MPB. Moreover, increasing the sintering time of BSF-PT samples which exhibit two dielectric anomalies does not result in samples with single maxima in ϵ_r and $\tan \delta$. This suggests that thermodynamic immiscibility of two different ferroelectric phases is the more likely cause of the relatively wider pseudo-MPB for $y=0.7$ and 0.8 and the lower piezoelectric activity. Preliminary energy dispersive x-ray spectroscopy (EDS) data did not reveal a significant difference in the composition of the tetragonal and rhombohedral regions. However, EDS data of Pb-based compositions are notoriously difficult with little accurate work performed due to PbO (and here Bi₂O₃) volatilization during irradiation. Further work is therefore necessary to clarify the extent of the compositional variation in these samples.

V. CONCLUSIONS

$(1-x)\text{BiSc}_{1-y}\text{Fe}_y\text{O}_3-x\text{PbTiO}_3$ ceramics have been fabricated in search of a commercially viable BiMeO₃–PbTiO₃ high T_C piezoelectric ceramic. 50% substitution of Sc by Fe yields single-phase ceramics with a narrow classical MPB (~ 2 mol%) similar to PZT. For $y=0.5$, MPB compositions exhibit high d_{33} (300 pC/N) and T_C (440 °C). Moreover, the electromechanical coupling coefficient ($k_p=0.49$) remains stable up to 250 °C, indicating that BSF-PT ceramics may be suitable for high temperature piezoelectric applications, although high field measurements at elevated temperatures are still required to establish unambiguously the viability of this system. Further substitution of Fe for Sc ($y=0.7$ and 0.8) causes the piezoelectric coefficient to deteriorate from 300 pC/N (50% substitution) to 170 pC/N and ~ 100 pC/N for $y=0.7$ and 0.8 , respectively. Increasing y promotes a widening of the region (~ 10 mol%) over which rhombohedral and tetragonal phases coexist as a result of thermodynamic immiscibility, creating a pseudo–rather than classical–MPB region. Within this pseudo-MPB region, the volume fraction of rhombohedral and tetragonal phases that have similar free energies is less than within a conventional MPB region.

Therefore, reorientation of the polarization vector between [001] and [111] is only possible in small regions of the sample, globally restricting domain wall motion and diminishing extrinsic contributions to the piezoactivity.

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